

Personalized Content Discovery Engine

**Cult Open Projects 2026
(AI/ML Track)**

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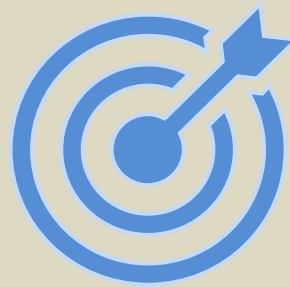
The Problem: Navigating the Sea of Content

The Challenge



Modern streaming platforms host massive content libraries
Users experience decision paralysis

The Goal



Build an intelligent system that learns individual user preferences and surfaces highly relevant content.

The Dataset



The Netflix Prize Dataset containing millions of ratings.

Data Exploration & Sparsity



98% Sparsity

Most users rate only a tiny fraction of movies, leaving the user–movie matrix nearly empty.



Rating Bias

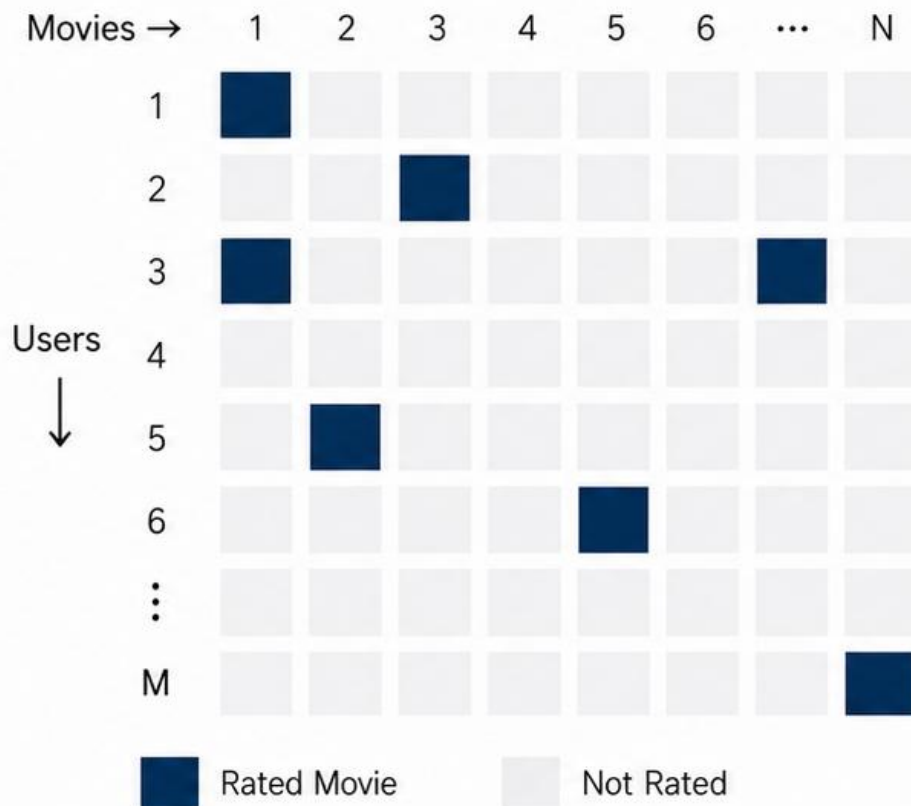
Ratings are concentrated around 3–5 stars, creating a strong positive skew.



Why It Matters

Traditional nearest-neighbor methods struggle because users rarely share enough common ratings.

USER–MOVIE INTERACTION MATRIX



98%

EMPTY MATRIX

Only 2%
interactions exist.

The Long Tail Phenomenon



Content Popularity

A tiny percentage of blockbuster films receive millions of ratings, while thousands of indie films receive almost none.



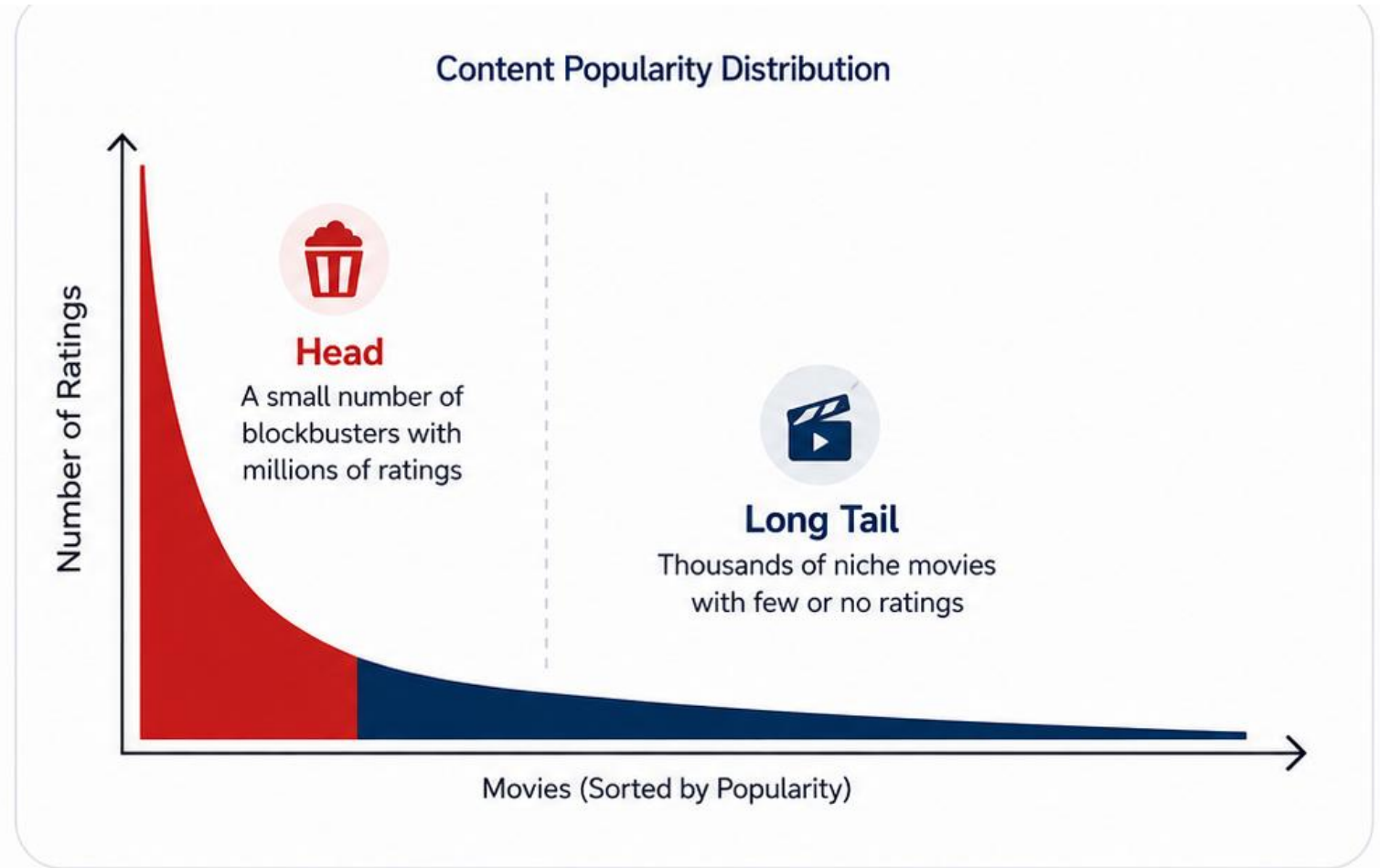
User Activity

A small group of power users generates the bulk of the data.



The Goal

A robust engine must look past blockbusters to help users discover the long tail of niche content.



Methodology & Architecture



Data Processing

Raw text data parsed into heavily optimized scipy sparse matrices to handle millions of interactions efficiently.



Train/Test Split

Data divided 80/20 to ensure models are evaluated on unseen user behavior.



The Algorithm

Transitioned from baseline Memory based Collaborative Filtering to advanced Model based Matrix Factorization.



Raw Text
Data



Scipy Sparse
Matrix



Train/Test
Split



Model
Training



User ratings,
movie info,
etc.

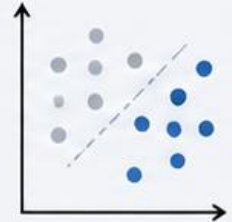


Optimized sparse
matrix for millions
of interactions



80%
Train Set

20%
Test Set



Model based
Matrix
Factorization

The Model: Singular Value Decomposition



How it Works

SVD breaks down the massive user-item grid into two smaller matrices: User-Features and Item-Features.



Latent Factors

It automatically discovers hidden themes (e.g., Action Heavy, Dark Comedy) without being explicitly programmed.



Prediction

By multiplying a user's hidden preferences by a movie's hidden traits, we can predict their exact rating.



Evaluation Metrics



1. RMSE

(Root Mean Squared Error)



Measures the mathematical accuracy of our predicted ratings against the real test data.

Lower is better.



2. MAP@10

(Mean Average Precision at 10)



Measures the ranking quality of our Top 10 recommendations.

We only rewarded the model if the recommended movie actually received a rating of 3.5 stars or higher.

Higher is better.

Future Roadmap



Hybrid Systems

Incorporating movie metadata (Genres, Directors) to solve the Cold Start problem for brand-new users.



Deep Learning

Exploring Neural Collaborative Filtering (NCF) to capture non-linear, complex behavioral patterns.



Explainable AI

Adding transparency features to the UI (e.g. "Recommended because you previously watched...") to build trust.

